

Hydra Methology

Drafting notes

Using organization of MS_SCAA methodology as a template

Also compare with Vanessa's multiple species state space model: https://github.com/vtrijoulet/Multisp_model_JAE (link to paper) https://github.com/vtrijoulet/MS_SSM (link to up to date model)

Model description

Population dynamics For an individual species i , the number of individuals N in size class j at time $t + 1$ represent the survivors from that cohort from the previous time step (t) that have not grown into a larger size class, plus the number of individuals growing into size class j from smaller size classes during the time interval. We only consider transitions among immediately adjacent size classes. For size classes larger than the recruit (smallest) size class, the model for each species i can be written:

$$N_{i,j,t+1} = \sum_{j'} \phi_{i,j' \rightarrow j,t} S_{i,j',t} N_{i,j',t}$$

The proportion surviving from size class $j - 1$ to size class j for a species at time t is denoted $S_{j',t}$. $\phi_{j' \rightarrow j}$ is the probability of moving from size class $j - 1$ to size class j .

[Lots more here but GF wanted to make progress on the objective function, see Gaichas et al. 2017 supplementary for rest of Hydra specs]

Recruitment Average recruitment with annual deviations

In the most basic formulation, no spawner-recruit model is specified, but annual log deviations $\Lambda_{i,t}$ from mean log recruitment $\bar{R}_i$ determine annual recruitment levels:

$$N_{i,1,t} = e^{\bar{R}_i + \Lambda_{i,t}}$$

Mortality The survival rate is:

$$S_{i,j,t} = e^{-(M1_{i,j} + M2_{i,j,t} + s_{i,j,q} F_{i,j,q,t})}$$

The rate of natural mortality is partitioned into two components: sources of natural mortality due to all unmodeled factors ($M1$) and predation mortality from species included in the model ($M2$).

The predation process is decomposed into size selectivity and species vulnerability in each area. The ρ , of prey species m size n for a given predator species i size j is:

$$\rho_{i,j,m,n} = \vartheta_{n,j} \lambda_{m,i}$$

where $\vartheta_{n,j}$ is the preference for a prey item of size n by predator size j , and $\lambda_{m,i}$ is the vulnerability of prey species m to predator species i . The parameter λ is set to either 0 or 1 depending on whether predator i is known to prey on species m .

The size preference function is:

$$\vartheta_{n,j} = \frac{1}{(w_n/w_j)\sigma_j\sqrt{2\pi}} e^{-\frac{[\log_e(w_n/w_j) - \Psi_j]^2}{2\sigma_j^2}}$$

where w_n is the weight at the midpoint of the length bin for a prey of size n , w_j is the weight at the midpoint of the length bin for a predator of size j , Ψ_j is the ‘preferred’ predator/prey weight ratio on a logarithmic scale, and σ_j^2 is the variance in predator size preference.

To estimate the daily food intake I for each predator, we use:

$$I_{i,j,t} = 24[\delta_i e^{\omega_i T}] \bar{C}_{i,j,k,t}$$

where the term in brackets gives the temperature (T)-dependent hourly consumption rate with parameters δ_i and ω_i and $\bar{C}_{i,j,k,t}$ is the mean stomach content weight (g) over a diel cycle.

The predation mortality rate on a particular prey species depends on the total consumption by predators, the suitability of that prey to each predator, and the total suitable prey biomass available to each of its predators. If the predation functional feeding response is assumed to be a Holling Type II, the $M2$ component on prey species m size n is then (Hall et al., 2006; Magnússon, 1995):

$$M2_{m,n,t} = \sum_i \sum_j I_{i,j,t} N_{i,j,t} \frac{\rho_{i,j,m,n}}{\sum_a \sum_b \rho_{i,j,a,b} W_{a,b} N_{a,b} + \Omega}$$

where a and b represent all prey species and sizes for predator i , W is the mean weight of prey a in size class b , Ω is ‘other’ food not explicitly included in the model, and all other terms are as above.

Fishing mortality is also size-specific. The selectivity for a specified gear type q is given by:

$$s_{i,j,q} = [1 + e^{-(c_{i,q} + d_{i,q}l)}]^{-1}$$

where $c_{i,j,q}$ and $d_{i,j,q}$ are model parameters and l is the midpoint of the j th length interval.

Fishing mortality F is currently defined as fishing effort multiplied by catchability for each fleet, summed across the three fleets.

The yield from species i at time t is then:

$$Y_{i,t} = \sum_j \sum_q \frac{s_{i,j,q} F_{i,j,q,t}}{M1_{i,j} + M2_{i,j,t} + s_{i,j,q} F_{i,j,q,t}} [1 - e^{-(M1_{i,j} + M2_{i,j,t} + s_{i,j,q} F_{i,j,q,t})}] N_{i,j,t} W_{i,j}$$

where $W_{i,j,k}$ is the mean weight of an individual of species i and length j .

Objective Function

The model can be fitted to multiple data streams. Currently the objective function comprises the following components, catch (by species for each fleet), size composition of the catch (by species for each fleet), survey abundance index (by species for each survey), survey size composition (by species for each survey), survey diet composition (stomach weight by predator size bin for each predator species and survey), and recruitment (annual deviations from mean recruitment by species).

Catch Expected catch for species i in fleet q is summed over time steps to the fishing/calendar year y .

$$\hat{C}_{y,i,q} = \sum_{t \in y} \sum_j \frac{s_{i,j,q} F_{i,j,q,t}}{M1_{i,j,t} + M2_{i,j,t} + s_{i,j,q} F_{i,j,q,t}} [1 - e^{-(M1_{i,j,t} + M2_{i,j,t} + s_{i,j,q} F_{i,j,q,t})}] N_{i,j,t} W_{i,j,t}$$

$$\ln(C_{y,i,q}) \sim \mathcal{N}(\ln(\hat{C}_{y,i,q}), \sigma_{q,i,y}^2)$$

Catch length composition

$$\hat{p}_{q,i,y,j} = p_{q,i,y,j}^* / \sum_{j'} p_{q,i,y,j'}^*$$

$$p_{q,i,y,j}^* = \sum_{t \in y} \frac{s_{i,j,q} F_{i,j,q,t}}{M1_{i,j,t} + M2_{i,j,t} + s_{i,j,q} F_{i,j,q,t}} \left[1 - e^{-(M1_{i,j,t} + M2_{i,j,t} + s_{i,j,q} F_{i,j,q,t})} \right] N_{i,j,t}$$

Proportions at length (in numbers) in catch for each species in each fleet are modeled as multinomial, given an input annual sample size for each composition.

Survey abundance indices The predicted values for survey abundance rely on the timing of the survey within a year, catchability of the species in the survey, and the (length-based) survey selectivity. Survey g predicted values (in weight) are:

$$\hat{I}_{g,i,t} = q_{i,g,t} \sum_j s_{i,j,g} N_{i,j,t} l_j W_{i,j,t}$$

The observed annual index values for each species are assumed to be lognormally distributed around the predicted values:

$$\ln(I_{g,i,t}) \sim \mathcal{N}(\ln(\hat{I}_{g,i,t}), \sigma_{g,i,t}^2)$$

Survey length composition

$$\hat{p}_{g,i,t,j} = p_{g,i,t,j}^* / \sum_{j'} p_{g,i,t,j'}^*$$

$$p_{g,i,t,j}^* = s_{i,j,g,t} N_{i,j,t}$$

Proportions at length (in numbers) in the survey for each species are modeled as multinomial, with an additional input sample size for each composition.

Prey proportions in survey diet Proportions by weight in stomach of predator i of size j of prey m :

$$\hat{\phi}_{g,t,i,j,m} = \phi_{g,t,i,j,m}^* / \left(\phi_{g,t,i,j,other}^* + \sum_m \phi_{g,t,i,j,m}^* \right)$$

$$\phi_{g,t,i,j,m}^* = \sum_n \rho_{i,j,m,n} N_{m,n,t} W_{m,n,t}$$

Proportions by prey species (in weight) in the survey are modeled as multinomial with an input sample size for each composition (potentially these data could be modeled as Dirichlet or delta-dirichlet, or...)

Penalties

Recruitment deviations A penalty to the objective function is added for the annual deviations from the stock-recruitment relationship for each species, which are assumed to be log-normal around the expected recruitment (equation 8):

$$\Lambda_{i,t} \sim \mathcal{N}(0, \tau_{\Lambda_i}^2)$$

where $\tau_{\Lambda_i}^2$ is the variance of the recruitment deviations for species i .